find minima using newton Raphson method

$f(x) = x^a - ax + b$ $\qquad a \to 3$ to 5

Take initial guess $= 2$ $\qquad b \to 5$ to 10

$$f'(x) = ax^{a-1} - a$$

$$f''(x) = a(a-1)x^{a-2}$$

Iteration 1 $\qquad [x_1 = 2]$

$$x_2 = x_1 - \frac{f'(x_1)}{f''(x_1)}$$

$x_2 = \square$

x_1	x_2	$f(x_2)$
2	✓	
⋮		

$x_1 = x_2 = \square$

Iteration 2

$$x_2 = x_1 - \frac{f'(x_1)}{f''(x_1)}$$

$x_2 = \square$